

On the extended W -algebra of type $\mathfrak{sl}_2$ at positive rational level

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Outline

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- 2 Overview of $\mathcal{M}_{p_+, p_-}$
- 3 Screening operators
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Review of VOA definitions

Vertex operator algebras (VOAs) are mathematically rigorous formulations of the chiral algebra of conformal field theories.

Def: A vertex operator algebra is a graded vector space

$$V = \bigoplus_{n \geq 0} V[n].$$

- $\dim V[n] < \infty$, $\dim V[0] = 1$.
- Vacuum vector $|0\rangle \in V[0]$ and conformal vector $T \in V[2]$.
- For $A \in V[h_A]$,

$$Y(A; z) = \sum_{n \in \mathbb{Z}} A_n z^{-n-h_A},$$

$$Y(T; z) = \sum_{n \in \mathbb{Z}} L_n z^{-n-2},$$

$$[L_m, L_n] = (m - n)L_{m+n} + \delta_{m+n,0} m(m^2 - 1) \frac{c_V}{12}.$$

Review of VOA definitions

VOAs satisfy the axioms:

- $Y(|0\rangle; z) = \text{id}.$
- $V[n] = \{A \in V \mid L_0 A = nA\}.$
- Operator state correspondence: $Y(A; z)|0\rangle = A + zV[[z]].$
- For $A \in V$, $\partial_z Y(A; z) = Y(L_{-1}A; z).$
- Locality: For $A, B \in V$ there exists an $N > 0$ such that $(z - w)^N[Y(A; z), Y(B; w)] = 0.$
- Operator product expansion: For $A \in V[h_A], B \in V$

$$\begin{aligned} Y(A; z)Y(B; w) &= Y(Y(A; z - w)B; w) \\ &= \sum_{n \in \mathbb{Z}} Y(A_n B; w)(z - w)^{-n - h_A}. \end{aligned}$$

Review of VOA definitions

Def: A V -module M is a $\mathbb{C}$ graded vector space

$$M = \bigoplus_{h \in \mathbb{C}} M[h]$$

that carries a representation $Y_M(\cdot; z)$ such that:

- The $M[h]$ are generalised L_0 eigenspaces with generalised eigenvalue h .
- For fixed $\text{im}(h)$, $\dim M[h] = 0$ for $\text{re}(h) \ll 0$
- For $A, B \in V$ and $C \in M$

$$Y_M(|0\rangle; z) = \text{id}_M$$

$$Y_M(A; z)C \in M((z))$$

$$Y_M(A; z)Y_M(B; w)C = Y_M(Y(A; z - w)B; w)C$$

Overview of $\mathcal{M}_{p_+, p_-}$

This talk is about the extended W -algebra of type $\mathfrak{sl}_2$ at positive rational level. We denote this VOA by $\mathcal{M}_{p_+, p_-}$.

- The VOA $\mathcal{M}_{p_+, p_-}$ is given by the kernels of 2 screening operators S_+, S_- on a lattice VOA $\mathcal{V}_{p_+, p_-}$.
- The central charge of $\mathcal{M}_{p_+, p_-}$ is the central charge of the minimal models

$$c_{p_+, p_-} = 1 - 6 \frac{(p_+ - p_-)^2}{p_+ p_-}$$

- As a module over itself $\mathcal{M}_{p_+, p_-}$ is reducible

$$0 \rightarrow \mathcal{X}_{1,1}^+ \longrightarrow \mathcal{M}_{p_+, p_-} \longrightarrow L(0, c_{p_+, p_-}) \rightarrow 0$$

- As a Virasoro module $\mathcal{M}_{p_+, p_-}$ decomposes into an ∞ direct sum of Virasoro modules. All but one of these modules are simple.

Overview of $\mathcal{M}_{p_+, p_-}$

- The $\mathcal{M}_{p_+, p_-}$ are the VOAs of certain logarithmic CFTs, that is, there exist $\mathcal{M}_{p_+, p_-}$ -modules on which the action of L_0 is non-diagonalisable.
- $\mathcal{M}_{2,3}$ is believed to be an extension of the dilute polymer and critical percolation statistical systems.

Overview of $\mathcal{M}_{p_+, p_-}$

The decomposition of $\mathcal{M}_{p_+, p_-}$ in terms of Virasoro highest weight vectors (also known as singular vectors) is

$$\begin{array}{ccccccc} & & & & |0\rangle & & \\ & & & & W_{0,0} & & \\ & & & W_{1,-1} & W_{1,0} & W_{1,1} & \\ & & W_{2,-2} & W_{2,-1} & W_{2,0} & W_{2,1} & W_{2,2} \\ \cdot & \cdot & \cdot & & & & \cdot & \cdot & \cdot \end{array}$$

Overview of $\mathcal{M}_{p_+, p_-}$

The screening operator formalism allows us to define Virasoro intertwiners, called screening operators, with which we can construct all Virasoro singular vectors in $\mathcal{M}_{p_+, p_-}$.

$$\begin{array}{ccccccc}
 & & & & & & |0\rangle \\
 & & & & & & S_+^{[p_+-1]}|\alpha_0^+\rangle \\
 & & & & & S_+^{[p_+-1]}|\alpha_1^+\rangle & S_+^{[2p_+-1]}|\alpha_1^+\rangle & S_+^{[3p_+-1]}|\alpha_1^+\rangle \\
 S_+^{[p_+-1]}|\alpha_2^+\rangle & S_+^{[2p_+-1]}|\alpha_2^+\rangle & S_+^{[3p_+-1]}|\alpha_2^+\rangle & S_+^{[4p_+-1]}|\alpha_2^+\rangle & S_+^{[5p_+-1]}|\alpha_2^+\rangle
 \end{array}$$

Overview of $\mathcal{M}_{p_+, p_-}$

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$$\begin{array}{ccccccc}
 & & & & & & |0\rangle \\
 & & & & & & S_-^{[p_- - 1]} |\alpha_0^- \rangle \\
 & & & & & & S_-^{[3p_- - 1]} |\alpha_1^- \rangle \quad S_-^{[2p_- - 1]} |\alpha_1^- \rangle \quad S_-^{[p_- - 1]} |\alpha_1^- \rangle \\
 & & & & & & S_-^{[5p_- - 1]} |\alpha_2^- \rangle \quad S_-^{[4p_- - 1]} |\alpha_2^- \rangle \quad S_-^{[3p_- - 1]} |\alpha_2^- \rangle \quad S_-^{[2p_- - 1]} |\alpha_2^- \rangle \quad S_-^{[p_- - 1]} |\alpha_2^- \rangle
 \end{array}$$

Overview of $\mathcal{M}_{p_+, p_-}$

Using the screening operator formalism we can define so-called Frobenius homomorphisms $E, F \in \text{End}(\mathcal{M}_{p_+, p_-})$, that satisfy

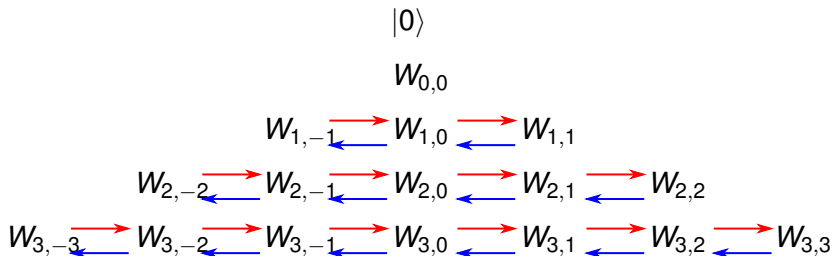
- $[E, L_n] = 0 = [F, L_n], n \in \mathbb{Z}$
- For $A \in \mathcal{M}_{p_+, p_-}$

$$[E, Y(A; z)] = Y(EA; z)$$

$$[F, Y(A; z)] = Y(FA; z)$$

Overview of $\mathcal{M}_{p_+, p_-}$

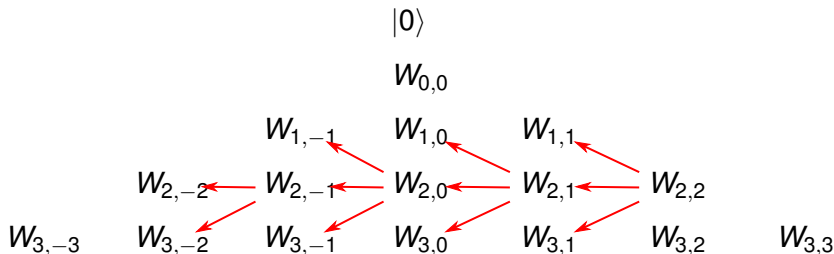
The action of the Frobenius homomorphisms E, F on $\mathcal{M}_{p_+, p_-}$ is given by



Overview of $\mathcal{M}_{p_+, p_-}$

The Frobenius operators allow us to prove the following:

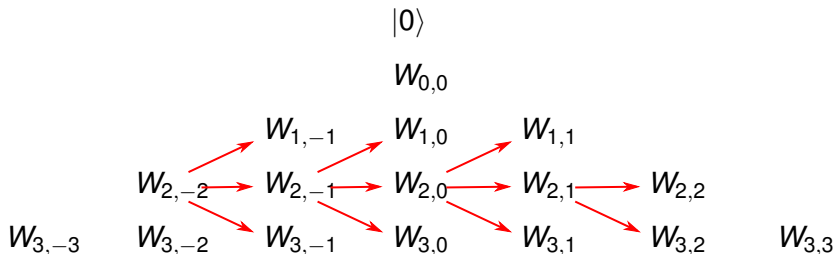
Thm The VOA $\mathcal{M}_{p_+, p_-}$ is generated by $T(z)$, $Y(W_{1,-1}; z)$, $Y(W_{1,1}; z)$



Overview of $\mathcal{M}_{p_+, p_-}$

The Frobenius operators allow us to prove the following:

Thm The VOA $\mathcal{M}_{p_+, p_-}$ is generated by $T(z)$, $Y(W_{1,-1}; z)$, $Y(W_{1,1}; z)$



Overview of $\mathcal{M}_{p_+, p_-}$

Knowledge of the generators of $\mathcal{M}_{p_+, p_-}$, the Frobenius operators and the screening operator formalism allows us to prove:

- Relations in Zhu's algebra $A_0(\mathcal{M}_{p_+, p_-})$.
- $\mathcal{M}_{p_+, p_-}$ satisfies Zhu's c_2 cofiniteness condition.
- $\mathcal{M}_{p_+, p_-}$ has exactly $(p_+ - 1)(p_- - 1)/2 + 2p_+p_-$ inequivalent simple modules.

The construction of $\mathcal{M}_{p_+, p_-}$

Plan:

- Define the free field (a.k.a. Heisenberg VOA) $\mathcal{F}_{\alpha_0}$
- Define an extension $\mathcal{V}_{p_+, p_-}$ of $\mathcal{F}_{\alpha_0}$ called a lattice VOA.
- Identify screening operators.
- Define $\mathcal{M}_{p_+, p_-}$ as a subVOA of $\mathcal{V}_{p_+, p_-}$ in terms of screening operators.

The construction of $\mathcal{M}_{p_+, p_-}$

Def Heisenberg VOA $\mathcal{F}_{\alpha_0}, \alpha_0 \in \mathbb{C}$

- Generated by free boson field $b(z) = \sum_{n \in \mathbb{Z}} b_n z^{-n-1}$

$$[b_m, b_n] = m\delta_{m, -n} \cdot 1$$

- $b(z)b(w) = \frac{1}{(z-w)^2} + \dots$
- Energy momentum tensor

$$T(z) = \frac{1}{2} : b(z)^2 : + \frac{\alpha_0}{2} \partial b(z)$$
$$c = 1 - 3\alpha_0^2$$

The construction of $\mathcal{M}_{p_+, p_-}$

Simple $\mathcal{F}_{\alpha_0}$ -modules are given by Fock modules F_λ , $\lambda \in \mathbb{C}$.

Def The Fock module F_λ , $\lambda \in \mathbb{C}$ is generated by a state $|\lambda\rangle$ such that

- $b_n|\lambda\rangle = \delta_{n,0}\lambda|\lambda\rangle$, $n \geq 0$
- $F_\lambda \cong \mathbb{C}[b_{-1}, b_{-2}, \dots]|\lambda\rangle$

Rmk $\mathcal{F}_{\alpha_0} \cong F_0$

Prop The category of $\mathcal{F}_{\alpha_0}$ -modules is semisimple.

- Simple objects are given by F_λ , $\lambda \in \mathbb{C}$
- The conformal weight of $|\lambda\rangle \in F_\lambda$ is

$$h_\lambda = \frac{1}{2}\lambda(\lambda - \alpha_0)$$

$$L_0|\lambda\rangle = h_\lambda|\lambda\rangle$$

Important parameters

The lattice VOA $\mathcal{V}_{p_+, p_-}$ is an extension of $\mathcal{F}_{\alpha_0}$ at special values of α_0 .
Let $p_+, p_- \geq 2$, $(p_+, p_-) = 1$

$$\alpha_+ = \sqrt{\frac{2p_-}{p_+}}, \quad \alpha_- = -\sqrt{\frac{2p_+}{p_-}}$$
$$\alpha = p_+ \alpha_+ = -p_- \alpha_- = \sqrt{2p_+ p_-}, \quad \alpha_0 = \alpha_+ + \alpha_-$$
$$\kappa_+ = \frac{\alpha_+^2}{2} = \frac{p_-}{p_+}, \quad \kappa_- = \frac{\alpha_-^2}{2} = \frac{p_+}{p_-}$$

Let Y be the lattice $Y = \mathbb{Z}\alpha = \mathbb{Z}\sqrt{2p_+p_-}$.

$$\phi(z) = \hat{b} + b_0 \log z - \sum_{n \neq 0} \frac{b_n}{n} z^{-n}, \quad [b_n, \hat{b}] = \delta_{n,0} 1, \quad \partial \phi(z) = b(z)$$

Def The lattice VOA $\mathcal{V}_{p_+, p_-}$ is given by

$$\mathcal{V}_{p_+, p_-} = \bigoplus_{\lambda \in Y} F_\lambda = \bigoplus_{n \in \mathbb{Z}} F_{n\alpha} \supset \mathcal{F}_{\alpha_0}$$

with fields $T(z)$ and $b(z)$ as in $\mathcal{F}_{\alpha_0}$, $Y(|\lambda|; z) = V_\lambda(z) =: e^{\lambda\phi(z)} :$

$$b(z) V_\lambda(w) = \frac{\lambda}{z-w} V_\lambda(w) + \dots$$

$$V_\lambda(z) V_\mu(w) = (z-w)^{\lambda\mu} : V_\lambda(z) V_\mu(w) :$$

$$T(z) V_\lambda(w) = \frac{\frac{1}{2}\lambda(\lambda - \alpha_0)}{(z-w)^2} V_\lambda(w) + \frac{1}{z-w} \partial V_\lambda(w) + \dots$$

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Let $Y = \mathbb{Z}\alpha = \mathbb{Z}\sqrt{2p_+p_-}$, $X = \text{Hom}_{\mathbb{Z}}(Y, \mathbb{Z}) = \mathbb{Z}\frac{1}{\alpha} = \mathbb{Z}\frac{1}{\sqrt{2p_+p_-}}$.

Prop

- The central charge of $\mathcal{V}_{p_+, p_-}$

$$c_{p_+, p_-} = 1 - 6 \frac{(p_+ - p_-)^2}{p_+ p_-}$$

is the same as for the minimal models.

- The category of $\mathcal{V}_{p_+, p_-}$ -modules is semisimple. There are $2p_+p_-$ simple objects.

$$V_{[\mu]} = \bigoplus_{\lambda \in \mu + Y} F_{\lambda}, \quad \mu \in X.$$

The simple modules of $\mathcal{V}_{p_+, p_-}$ are parametrised by the classes of X/Y .

Recall that $\alpha_{\pm} = \pm\sqrt{2p_{\mp}/p_{\pm}} \in X$. Since $h_{\alpha_{\pm}} = 1$, $V_{\alpha_{\pm}}(w) = S_{\pm}(w)$ satisfies

$$T(z)S_{\pm}(w) = \frac{1}{(z-w)^2}S_{\pm}(w) + \frac{1}{z-w}\partial S_{\pm}(w) + \dots$$

Prop The $S_{\pm}(z)$ form screening operators, that is,

$$\begin{aligned} S_{\pm} &= \text{Res}_{w=0} S_{\pm}(w)dw : \mathcal{V}_{p_+,p_-} \rightarrow V_{[\alpha_{\pm}]} \\ [L_n, S_{\pm}] &= 0, \quad n \in \mathbb{Z} \end{aligned}$$

Def The VOA $\mathcal{M}_{p_+,p_-}$ is given by

$$\mathcal{M}_{p_+,p_-} = \ker(S_+ : \mathcal{V}_{p_+,p_-} \rightarrow V_{[\alpha_+]}) \cap \ker(S_- : \mathcal{V}_{p_+,p_-} \rightarrow V_{[\alpha_-]})$$

Summary and next steps

Summary:

- The VOA $\mathcal{M}_{p_+,p_-}$ is a subVOA of the lattice VOA $\mathcal{V}_{p_+,p_-}$.
- We have constructed $\mathcal{M}_{p_+,p_-}$ as the intersection of kernels of screening operators.

Next steps:

- In order to understand $\mathcal{M}_{p_+,p_-}$ better, we need the screening operator formalism.
- The screening operator formalism involves integrating products of screening operators over certain cycles. Determining these cycles at the minimal model central charge $c_{p_+,p_-} = 1 - 6(p_+ - p_-)^2/p_+p_-$ is a very hard problem.
- By deforming the central charge in a deformation parameter ε , one can work at generic central charge.
- By setting $\varepsilon = 0$ one can specialise results obtained at generic central charge.

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Next steps:

- In order to understand $\mathcal{M}_{p_+, p_-}$ better, we need the screening operator formalism.
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- By deforming the central charge in a deformation parameter ε , one can work at generic central charge.
- By setting $\varepsilon = 0$ one can specialise results obtained at generic central charge.

Deforming the conformal structure

Let $\mathcal{O} = \mathbb{C}[[\varepsilon]]$, $\mathcal{K} = \mathbb{C}((\varepsilon))$. We deform our previous parameters

$$\alpha_+(\varepsilon) = \sqrt{\frac{2p_-}{p_+}} + \alpha_+^{(1)}\varepsilon + \alpha_+^{(2)}\varepsilon^2 + \dots \in \mathcal{O} \quad \alpha_+^{(1)} \neq 0$$

$$\alpha_-(\varepsilon) = -\sqrt{\frac{2p_+}{p_-}} + \alpha_-^{(1)}\varepsilon + \alpha_-^{(2)}\varepsilon^2 + \dots \in \mathcal{O} \quad \alpha_-^{(1)} \neq 0$$

$$\kappa_{\pm}(\varepsilon) = \frac{\alpha_{\pm}(\varepsilon)^2}{2}$$

such that

$$\alpha_+(\varepsilon)\alpha_-(\varepsilon) = -2 \quad \alpha_0(\varepsilon) = \alpha_+(\varepsilon) + \alpha_-(\varepsilon)$$

$$\kappa_+(\varepsilon)\kappa_-(\varepsilon) = 1$$

Deforming the conformal structure

Plan:

- Define the Heisenberg VOA and Fock modules over $\mathcal{K}$ and $\mathcal{O}$.
- Identify screening operators.
- Define integrals of products of screening operators over $\mathcal{K}$.
- Check that these are also well defined over $\mathcal{O}$.
- Specialise to $\varepsilon = 0$ and obtain screening operators over $\mathbb{C}$ at central charge c_{p_+, p_-} .

Deforming the conformal structure

Def Heisenberg VOA ${}_{\mathcal{K}}\mathcal{F}_{\alpha_0(\varepsilon)}$

- Generated by free boson field $b(z) = \sum_{n \in \mathbb{Z}} b_n z^{-n-1}$

$$[b_m, b_n] = m\delta_{m, -n} \cdot 1$$

- $b(z)b(w) = \frac{1}{(z-w)^2} + \dots$
- Energy momentum tensor

$$T(z) = \frac{1}{2} : b(z)^2 : + \frac{\alpha_0(\varepsilon)}{2} \partial b(z)$$
$$c = 1 - 3\alpha_0(\varepsilon)^2$$

Deforming the conformal structure

Simple ${}_{\mathcal{K}}\mathcal{F}_{\alpha_0(\varepsilon)}$ -modules are given by Fock modules ${}_{\mathcal{K}}F_{\lambda}, \lambda \in \mathcal{K}$.

Def The Fock module ${}_{\mathcal{K}}F_{\lambda}, \lambda \in \mathcal{K}$ is generated by a state $|\lambda\rangle$ such that

- $b_n|\lambda\rangle = \delta_{n,0}\lambda|\lambda\rangle, n \geq 0$
- As a $\mathcal{K}$ vector space ${}_{\mathcal{K}}F_{\lambda} \cong \mathcal{K}[b_{-1}, b_{-2}, \dots]|\lambda\rangle$

Rmk ${}_{\mathcal{K}}\mathcal{F}_{\alpha_0(\varepsilon)} \cong {}_{\mathcal{K}}F_0$

Prop The category of ${}_{\mathcal{K}}\mathcal{F}_{\alpha_0(\varepsilon)}$ -modules is semisimple.

- Simple objects are given by ${}_{\mathcal{K}}F_{\lambda}, \lambda \in \mathcal{K}$
- The conformal weight of $|\lambda\rangle \in {}_{\mathcal{K}}F_{\lambda}$ is

$$h_{\lambda} = \frac{1}{2}\lambda(\lambda - \alpha_0(\varepsilon))$$
$$L_0|\lambda\rangle = h_{\lambda}|\lambda\rangle$$

Deforming the conformal structure

For $\lambda \in \mathcal{O} = \mathbb{C}[[\varepsilon]]$, we can restrict to Fock modules over $\mathcal{O}$

$${}_{\mathcal{O}}F_{\lambda} = \mathcal{O}[b_{-1}, b_{-2}, \dots]|\lambda\rangle \subset {}_{\mathcal{K}}F_{\lambda}.$$

The Fock module ${}_{\mathcal{O}}F_0$ carries the structure of a VOA over $\mathcal{O}$.

Deforming the conformal structure

As in the undeformed case $h_{\alpha_{\pm}}(\varepsilon) = 1$ and the fields

$$S_{\pm}(z) =: e^{\alpha_{\pm}(\varepsilon)\phi(z)} :$$

define screening operators of ${}_{\mathcal{K}}F_0$ and ${}_{\mathcal{O}}F_0$.

Let ${}_{\mathcal{O}}X = \mathbb{Z}\frac{\alpha_+(\varepsilon)}{2} \oplus \mathbb{Z}\frac{\alpha_-(\varepsilon)}{2}$ be the rank 2 lattice parametrised by

$$\beta_{r,s}(\varepsilon) = \frac{1-r}{2}\alpha_+(\varepsilon) + \frac{1-s}{2}\alpha_-(\varepsilon), \quad r, s \in \mathbb{Z}.$$

Prop There exists a construction of cycles $[\Gamma_r(\kappa_+(\varepsilon))], [\Gamma_s(\kappa_-(\varepsilon))]$ over which one can integrate $S_{\pm}(z)$ such that

$$S_+^{[r]} = \int_{[\Gamma_r(\kappa_+(\varepsilon))]} S_+(z_1) \cdots S_+(z_r) dz_1 \cdots dz_r : {}_{\mathcal{K}}F_{\beta_{r,s}(\varepsilon)} \rightarrow {}_{\mathcal{K}}F_{\beta_{-r,s}(\varepsilon)}$$

$$S_-^{[s]} = \int_{[\Gamma_s(\kappa_-(\varepsilon))]} S_-(z_1) \cdots S_-(z_s) dz_1 \cdots dz_s : {}_{\mathcal{K}}F_{\beta_{r,s}(\varepsilon)} \rightarrow {}_{\mathcal{K}}F_{\beta_{r,-s}(\varepsilon)}$$

are non-trivial Virasoro homomorphisms.

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The S_+ case in greater detail

Let

$$U_r(z; \kappa_+(\varepsilon)) = \prod_{1 \leq i \neq j \leq r} (z_i - z_j)^{\kappa_+(\varepsilon)} \prod_{i=1}^r z_i^{(1-r)\kappa_+(\varepsilon)} = \prod_{1 \leq i \neq j \leq r} \left(1 - \frac{z_i}{z_j}\right)^{\kappa_+(\varepsilon)}$$

Then

$$\begin{aligned} S_+(z_1) \cdots S_+(z_r) dz_1 \cdots dz_r \Big|_{\mathcal{K}_{\beta_r, s(\varepsilon)}} &= e^{r\alpha_+(\varepsilon)\hat{b}} U_r(z; \kappa_+(\varepsilon)) \\ &\times \prod_{i=1}^r z_i^s \prod_{n \geq 1} e^{\alpha_+(\varepsilon) \sum_{i=1}^r \frac{z_i^n}{n} b_{-n}} \prod_{n \geq 1} e^{-\alpha_+(\varepsilon) \sum_{i=1}^r \frac{z_i^{-n}}{n} b_n} \frac{dz_1 \cdots dz_r}{z_1 \cdots z_r} \end{aligned}$$

The S_+ case in greater detail

The cycles $[\Gamma_r(\kappa_+(\varepsilon))]$ are normalised such that

$$\int_{[\Gamma_r(\kappa_+(\varepsilon))]} U_r(z; \kappa_+(\varepsilon)) \frac{dz_1 \cdots dz_r}{z_1 \cdots z_r} = 1.$$

Evaluating

$$S_+^{[r]} = \int_{[\Gamma_r(\kappa_+(\varepsilon))]} S_+(z_1) \cdots S_+(z_r) dz_1 \cdots dz_r : \mathcal{K} F_{\beta_r, s(\varepsilon)} \rightarrow \mathcal{K} F_{\beta_{-r}, s(\varepsilon)}$$

requires one to evaluate integrals of the form

$$(f, g)_{\kappa_+(\varepsilon)}^r := \int_{[\Gamma_r(\kappa_+(\varepsilon))]} U_r(z; \kappa_+(\varepsilon)) f(z_1^{-1}, \dots, z_r^{-1}) g(z_1, \dots, z_r) \frac{dz_1 \cdots dz_r}{z_1 \cdots z_r}$$

$$f, g \in {}_{\mathcal{K}}\Lambda_r = \mathcal{K}[z_1, \dots, z_r]^{S_r}$$

The S_+ case in greater detail

The cycles $[\Gamma_r(\kappa_+(\varepsilon))]$ are normalised such that

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Evaluating

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$$f, g \in \mathcal{K} \Lambda_r = \mathcal{K}[z_1, \dots, z_r]^{S_r}$$

Evaluating the action of $S_+^{[r]}$

Prop The pairing

$$(f, g)_{\kappa_+(\varepsilon)}^r = \int_{[\Gamma_r(\kappa_+(\varepsilon))]} U_r(z; \kappa_+(\varepsilon)) f(z_1^{-1}, \dots, z_r^{-1}) g(z_1, \dots, z_r) \frac{dz_1 \dots dz_r}{z_1 \dots z_r}$$

forms a non-degenerate symmetric inner product on

$${}_{\mathcal{K}}\Lambda_r = \mathcal{K}[z_1, \dots, z_r]^{S_r}.$$

Prop There exists a basis $P_\lambda(z; \kappa_+(\varepsilon))$ of ${}_{\mathcal{K}}\Lambda_r$ (and ${}_{\mathcal{O}}\Lambda_r$) parametrised by partitions of integers of length at most r , such that

$$(P_\lambda(z; \kappa_+(\varepsilon)), P_\mu(z; \kappa_+(\varepsilon)))_{\kappa_+(\varepsilon)}^r = 0, \text{ if } \lambda \neq \mu$$

$$(P_\lambda(z; \kappa_+(\varepsilon)), P_\lambda(z; \kappa_+(\varepsilon)))_{\kappa_+(\varepsilon)}^r \in \mathcal{O}^\times$$

This basis consists of Jack polynomials.

Evaluating the action of $S_+^{[r]}$

Prop The pairing

$$(f, g)_{\kappa_+(\varepsilon)}^r = \int_{[\Gamma_r(\kappa_+(\varepsilon))]} U_r(z; \kappa_+(\varepsilon)) f(z_1^{-1}, \dots, z_r^{-1}) g(z_1, \dots, z_r) \frac{dz_1 \dots dz_r}{z_1 \dots z_r}$$

forms a non-degenerate symmetric inner product on

$${}_{\mathcal{K}}\Lambda_r = \mathcal{K}[z_1, \dots, z_r]^{S_r}.$$

Prop There exists a basis $P_\lambda(z; \kappa_+(\varepsilon))$ of ${}_{\mathcal{K}}\Lambda_r$ (and ${}_{\mathcal{O}}\Lambda_r$) parametrised by partitions of integers of length at most r , such that

$$(P_\lambda(z; \kappa_+(\varepsilon)), P_\mu(z; \kappa_+(\varepsilon)))_{\kappa_+(\varepsilon)}^r = 0, \text{ if } \lambda \neq \mu$$

$$(P_\lambda(z; \kappa_+(\varepsilon)), P_\lambda(z; \kappa_+(\varepsilon)))_{\kappa_+(\varepsilon)}^r \in \mathcal{O}^\times$$

This basis consists of Jack polynomials.

Prop

$$S_+^{[r]} : \mathcal{O}_{F_{\beta_{r,s}(\varepsilon)}} \rightarrow \mathcal{O}_{F_{\beta_{-r,s}(\varepsilon)}}, \quad r \geq 1, s \in \mathbb{Z}$$

$$S_-^{[s]} : \mathcal{O}_{F_{\beta_{r,s}(\varepsilon)}} \rightarrow \mathcal{O}_{F_{\beta_{r,-s}(\varepsilon)}}, \quad s \geq 1, r \in \mathbb{Z}$$

Therefore, we can specialise to $\varepsilon = 0$. Let

$$\beta_{r,s} = \frac{1-r}{2}\alpha_+ + \frac{1-s}{2}\alpha_- \in X = \mathbb{Z}\frac{1}{\alpha}, \quad r, s \in \mathbb{Z}$$

Prop

$$S_+^{[r]} : F_{\beta_{r,s}} \rightarrow F_{\beta_{-r,s}}, \quad r \geq 1, s \in \mathbb{Z}$$

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Previously (= without deformation) these screening operators could only be defined for $1 \leq r < p_+$ or $1 \leq s < p_-$ respectively.

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Previously (= without deformation) these screening operators could only be defined for $1 \leq r < p_+$ or $1 \leq s < p_-$ respectively.

Since $\beta_{r,s} = \beta_{r+p_+,s+p_-}$, the previous proposition gives us an infinite list of Virasoro highest weight vectors (called singular vectors) in $F_{\beta_{r,s}}$. Let

$$\beta_{r,s;n} = \beta_{r-np_+,s} = \beta_{r,s+np_-}, \quad 1 \leq r \leq p_+, 1 \leq s \leq p_-, n \in \mathbb{Z}$$

Thm Let $1 \leq r \leq p_+, 1 \leq s \leq p_-, n \in \mathbb{Z}$, then all singular vectors of $F_{\beta_{r,s;n}}$ are given by

$$S_+^{[(k+n+1)p_+-r]} |\beta_{p_+-r,s;-1-2k-n}\rangle \propto S_-^{[(k+1)p_-s]} |\beta_{r,p_-s;2k+n+1}\rangle,$$

for $n \geq 0, k \geq 0$,

$$S_+^{[(k+1)p_+-r]} |\beta_{p_+-r,s;-1-2k+n}\rangle \propto S_-^{[(k-n+1)p_-s]} |\beta_{r,p_-s;2k-n+1}\rangle,$$

for $n \leq 0, k \geq 0$.

These formulae for singular vectors reproduce the singular vector decomposition of $\mathcal{M}_{p_+, p_-}$ from before:

$$\begin{array}{ccccccc}
 & & & & & & |0\rangle \\
 & & & & & & S_+^{[p_+-1]}|\alpha_0^+\rangle \\
 & & & & & S_+^{[p_+-1]}|\alpha_1^+\rangle & S_+^{[2p_+-1]}|\alpha_1^+\rangle & S_+^{[3p_+-1]}|\alpha_1^+\rangle \\
 S_+^{[p_+-1]}|\alpha_2^+\rangle & S_+^{[2p_+-1]}|\alpha_2^+\rangle & S_+^{[3p_+-1]}|\alpha_2^+\rangle & S_+^{[4p_+-1]}|\alpha_2^+\rangle & S_+^{[5p_+-1]}|\alpha_2^+\rangle
 \end{array}$$

These formulae for singular vectors reproduce the singular vector decomposition of $\mathcal{M}_{p_+, p_-}$ from before:

$$\begin{aligned}
 & |0\rangle \\
 & S_-^{[p_- - 1]} |\alpha_0^-\rangle \\
 & S_-^{[3p_- - 1]} |\alpha_1^-\rangle \quad S_-^{[2p_- - 1]} |\alpha_1^-\rangle \quad S_-^{[p_- - 1]} |\alpha_1^-\rangle \\
 & S_-^{[5p_- - 1]} |\alpha_2^-\rangle \quad S_-^{[4p_- - 1]} |\alpha_2^-\rangle \quad S_-^{[3p_- - 1]} |\alpha_2^-\rangle \quad S_-^{[2p_- - 1]} |\alpha_2^-\rangle \quad S_-^{[p_- - 1]} |\alpha_2^-\rangle
 \end{aligned}$$

Construction of simple $\mathcal{M}_{p_+, p_-}$ -modules

We parametrise the $2p_+p_-$ simple $\mathcal{V}_{p_+, p_-}$ -modules as

$$V_{r,s}^+ = \bigoplus_{n \in \mathbb{Z}} F_{\beta_{r,s;2n}}, \quad V_{r,s}^- = \bigoplus_{n \in \mathbb{Z}} F_{\beta_{r,s;2n+1}}, \quad 1 \leq r \leq p_+, 1 \leq s \leq p_-.$$

The screening operators $S_+^{[r]}, S_-^{[s]}$ define maps

$$S_+^{[r]} : V_{r,s}^\pm \rightarrow V_{p_+-r,s}^\mp, \quad 1 \leq r < p_+, 1 \leq s \leq p_-$$

$$S_-^{[s]} : V_{r,s}^\pm \rightarrow V_{r,p_--s}^\mp, \quad 1 \leq r \leq p_+, 1 \leq s < p_-$$

Prop For $A \in \mathcal{M}_{p_+, p_-}$, $1 \leq r < p_+, 1 \leq s < p_-$

$$[S_+^{[r]}, Y(A; z)] = 0 = [S_-^{[s]}, Y(A; z)].$$

Thus the kernels and images of $S_+^{[r]}, S_-^{[s]}$ are $\mathcal{M}_{p_+, p_-}$ -modules.

Let $1 \leq r < p_+, 1 \leq s < p_-$

$$\mathcal{K}_{r,s}^+ = \ker(\mathcal{S}_+^{[r]} : V_{r,s}^+ \rightarrow V_{p_+-r,s}^\mp) \cap \ker(\mathcal{S}_-^{[s]} : V_{r,s}^\pm \rightarrow V_{r,p_--s}^\mp)$$

Prop The following $\frac{1}{2}(p_+ - 1)(p_- - 1) + 2p_+p_-$ inequivalent $\mathcal{M}_{p_+,p_-}$ -modules are simple.

$$\mathcal{X}_{r,s}^\pm = \text{im}(\mathcal{S}_+^{[p_+-r]} : V_{p_+-r,s}^\mp \rightarrow V_{r,s}^\pm) \cap \text{im}(\mathcal{S}_-^{[p_--s]} : V_{r,p_--s}^\mp \rightarrow V_{r,s}^\pm)$$

$$\mathcal{X}_{r,p_-}^\pm = \text{im}(\mathcal{S}_+^{[p_+-r]} : V_{p_+-r,p_-}^\mp \rightarrow V_{r,p_-}^\pm)$$

$$\mathcal{X}_{p_+,s}^\pm = \text{im}(\mathcal{S}_-^{[p_--s]} : V_{p_+,p_--s}^\mp \rightarrow V_{p_+,s}^\pm)$$

$$\mathcal{X}_{p_+,p_-}^\pm = V_{p_+,p_-}^\pm$$

$$L_{[r,s]} = L_{[p_+-r,p_--s]} = \mathcal{K}_{r,s}^+ / \mathcal{X}_{r,s}^+$$

Thm The above $\frac{1}{2}(p_+ - 1)(p_- - 1) + 2p_+p_-$ modules form all isomorphism classes of simple $\mathcal{M}_{p_+,p_-}$ -modules.

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$$\mathcal{X}_{r,p_-}^\pm = \text{im}(\mathcal{S}_+^{[p_+-r]} : V_{p_+-r,p_-}^\mp \rightarrow V_{r,p_-}^\pm)$$

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Thm The above $\frac{1}{2}(p_+ - 1)(p_- - 1) + 2p_+p_-$ modules form all isomorphism classes of simple $\mathcal{M}_{p_+,p_-}$ -modules.

Diagrammatically the singular vector decomposition of the $\mathcal{X}_{r,s}^\pm$ is given by:



The construction of Frobenius operators

The screening operators $S_+^{[r]}$, $S_-^{[s]}$ arrange the lattice modules $V_{r,s}^\pm$ into complexes

$$\begin{aligned} \cdots \rightarrow V_{p_+-r,s}^\mp &\xrightarrow{S_+^{[p_+-r]}} V_{r,s}^\pm \xrightarrow{S_+^{[r]}} V_{p_+-r,s}^\mp \xrightarrow{S_+^{[p_+-r]}} V_{r,s}^\pm \rightarrow \cdots \\ \cdots \rightarrow V_{r,p_--s}^\mp &\xrightarrow{S_-^{[p_--s]}} V_{r,s}^\pm \xrightarrow{S_-^{[s]}} V_{r,p_--s}^\mp \xrightarrow{S_-^{[p_--s]}} V_{r,s}^\pm \rightarrow \cdots \end{aligned}$$

$$S_+^{[p_+-r]} \circ S_+^{[r]} = 0 = S_-^{[p_--s]} \circ S_-^{[s]},$$

called Felder complexes.

The construction of Frobenius operators

- The Frobenius operators are regularisations of $S_+^{[p_+-r]} \circ S_+^{[r]}, S_-^{[p_--s]} \circ S_-^{[s]}$ such that they are non-trivial.
- These regularisations are only well defined on $\ker S_+^{[r]}, \ker S_-^{[s]}$.

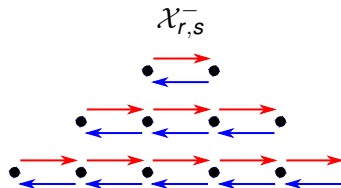
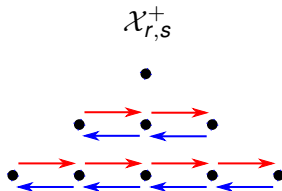
Thm The Frobenius operators are families of maps $E, F \in \text{End}(\mathcal{M}_{p_+,p_-}), E, F \in \text{End}(\mathcal{X}_{r,s}^\pm)$ that satisfy

- $[L_n, E] = 0 = [L_n, F], n \in \mathbb{Z}$
- For $A \in \mathcal{M}_{p_+,p_-}$

$$[E, Y(A; z)] = Y(EA; z)$$

$$[F, Y(A; z)] = Y(FA; z)$$

The action of E and F on the simple modules $\mathcal{X}_{r,s}^\pm$ is given by:



Sketch of construction of E

We work over $\mathcal{O} = \mathbb{C}[[\varepsilon]]$ and $\mathcal{K} = \mathbb{C}((\varepsilon))$ again. Let $1 \leq r < p_+$, $s \in \mathbb{Z}$

$$E = \int_{[\Gamma_{p_+-r}^\infty(\kappa_+(\varepsilon))]} \int_{[\Gamma_r(\kappa_+(\varepsilon))]} S_+(z_1) \cdots S_+(z_{p_+-r}) \\ \times S_+(w_1) \cdots S_+(w_r) dz_1 \cdots dz_{p_+-r} dw_1 \cdots dw_r : \mathcal{K} F_{\beta_r, s(\varepsilon)} \rightarrow \mathcal{K} F_{\beta_{r-2p_+}, s(\varepsilon)}$$

where the $[\Gamma_r(\kappa_+(\varepsilon))]$ are the cycles used to define $S_+^{[r]}$ and the cycle $[\Gamma_{p_+-r}^\infty(\kappa_+(\varepsilon))]$ is supported in the domain $|z_i| > |w_j|$.

Prop When restricted to ${}_{\mathcal{O}}F_{\beta_r, s(\varepsilon)}$, E defines a map

$$E : {}_{\mathcal{O}}F_{\beta_r, s(\varepsilon)} \rightarrow \varepsilon^{-1} {}_{\mathcal{O}}F_{\beta_{r-2p_+}, s(\varepsilon)}.$$

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Sketch of construction of E

Let

$${}_{\mathcal{O}}K_{r,s} = \{v \in {}_{\mathcal{O}}F_{\beta_{r,s}(\varepsilon)} \mid S_+^{[r]}v \in \varepsilon {}_{\mathcal{O}}F_{\beta_{-r,s}(\varepsilon)}\}.$$

${}_{\mathcal{O}}K_{r,s}$ specialises to $\ker S_+^{[r]}$ when $\varepsilon = 0$.

Prop On ${}_{\mathcal{O}}K_{r,s}$, E defines a map

$$E : {}_{\mathcal{O}}K_{r,s} \rightarrow {}_{\mathcal{O}}F_{\beta_{r-2p_+,s}(\varepsilon)}.$$

Therefore E is well defined over $\mathbb{C}$, when specialising to $\varepsilon = 0$. The remaining properties of E can be shown to hold modulo ε by explicit calculations involving Jack polynomials.

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How everything comes together

- Zhu's algebra $A_0(V)$ is an associative $\mathbb{C}$ algebra that can be constructed from a given VOA V .
- Zhu's algebra encodes the action of zero modes on highest weight states.
- There exists a 1-1 correspondence between finite-dimensional simple $A_0(V)$ -modules and simple V -modules.
- We can use our realisation of singular vectors to calculate the action of the zero modes of the corresponding fields on highest weight states.
- Our realisations of singular vectors therefore allow us to calculate relations in Zhu's algebra.

Eigenvalues on highest weight states

We can explicitly realise all $W_{n,0} \in \mathcal{M}_{p_+,p_-}$ as

$$W_{n,0} = S_+^{[(n+1)p_+-1]} |\beta_{p_+-1,1;-2n-1}\rangle.$$

This allows us to compute the eigenvalues of the zero modes $W_{n,0}[0]$ on $|\beta\rangle \in F_\beta, \beta \in \mathbb{C}$.

Prop For $n \geq 0$

$$\omega_n(\beta) = \langle \beta | W_{n,0}[0] | \beta \rangle \propto \prod_{i=1}^{(n+1)p_+-1} \prod_{j=1}^{(n+1)p_--1} (\beta - \beta_{i,j}).$$

Recall that $h_\beta = \frac{1}{2}\beta(\beta - \alpha_0)$. There exist polynomials $g_n \in \mathbb{C}[h]$, such that for $n \geq 0$

$$g_n(h_\beta) = \begin{cases} \omega_n(\beta) & n \text{ even} \\ \omega_n(\beta)^2 & n \text{ odd} \end{cases}$$

Eigenvalues on highest weight states

Let $\Delta_{r,s}$ be the conformal weights of the $L_{[r,s]}$ and $\Delta_{r,s}^{\pm}$ the conformal weights of the $\mathcal{X}_{r,s}^{\pm}$.

$$g_0(h_{\beta}) = \omega_0(\beta) \propto \prod_{[i,j]} (h_{\beta} - \Delta_{i,j}), \quad [i,j] \cong [p_+ - i, p_- - j]$$

$$g_1(h_{\beta}) = \omega_1(\beta)^2 \propto g_0(h_{\beta})^4 \prod_{i=1}^{p_+-1} \prod_{j=1}^{p_--1} (h_{\beta} - \Delta_{i,j}^+)^2$$

$$\prod_{i=1}^{p_+-1} (h_{\beta} - \Delta_{i,p_-}^+)^2 \prod_{j=1}^{p_--1} (h_{\beta} - \Delta_{p_+,j}^+)^2$$

$$(h_{\beta} - \Delta_{p_+,p_-}^+)$$

Eigenvalues on highest weight states

Let $\Delta_{r,s}$ be the conformal weights of the $L_{[r,s]}$ and $\Delta_{r,s}^{\pm}$ the conformal weights of the $\mathcal{X}_{r,s}^{\pm}$.

$$\begin{aligned}
 g_2(h_{\beta}) = \omega_2(\beta) &\propto \prod_{[i,j]} (h_{\beta} - \Delta_{i,j})^3 \\
 &\prod_{i=1}^{p_+-1} \prod_{j=1}^{p_--1} (h_{\beta} - \Delta_{i,j}^+)^2 \prod_{i=1}^{p_+-1} \prod_{j=1}^{p_--1} (h_{\beta} - \Delta_{i,j}^-) \\
 &\prod_{i=1}^{p_+-1} (h_{\beta} - \Delta_{i,p_-}^+)^2 \prod_{i=1}^{p_+-1} (h_{\beta} - \Delta_{i,p_-}^-) \\
 &\prod_{j=1}^{p_--1} (h_{\beta} - \Delta_{p_+,j}^+)^2 \prod_{j=1}^{p_--1} (h_{\beta} - \Delta_{p_+,j}^-) \\
 &(h_{\beta} - \Delta_{p_+,p_-}^+)(h_{\beta} - \Delta_{p_+,p_-}^-)
 \end{aligned}$$

The structure of Zhu's algebra $A_0(\mathcal{M}_{p_+,p_-})$

Thm Zhu's algebra $A_0(\mathcal{M}_{p_+,p_-})$ of $\mathcal{M}_{p_+,p_-}$ is a finite-dimensional $\mathbb{C}$ algebra.

- The Frobenius operators E, F are derivations of $A_0(\mathcal{M}_{p_+,p_-})$.
- $A_0(\mathcal{M}_{p_+,p_-})$ is generated by $[T], [W_{1,m}], m = -1, 0, 1$.
- $[T]$ satisfies the polynomial relations

$$\begin{aligned}g_0([T]) &= [W_{0,0}], \quad g_1([T]) = [W_{1,0}]^2 \\g_2([T]) &= [W_{2,0}] = 0\end{aligned}$$

The structure of Zhu's algebra $A_0(\mathcal{M}_{p_+, p_-})$

Thm (continued)

- The $[W_{1,m}]$, $m = -1, 0, 1$ satisfy the following multiplication formulae

$\cdot$	$[W_{1,-1}]$	$[W_{1,0}]$	$[W_{1,1}]$
$[W_{1,-1}]$	0	$-f([T])[W_{1,-1}]$	$-g_1([T]) - f([T])[W_{1,0}]$
$[W_{1,0}]$	$f([T])[W_{1,-1}]$	$g_1([T])$	$-f([T])[W_{1,1}]$
$[W_{1,1}]$	$-g_1([T]) + f([T])[W_{1,0}]$	$f([T])[W_{1,1}]$	0

where f is a polynomial of degree less than $(2p_+ - 1)(2p_- - 1)/2$. The polynomial f is non-trivial and the conformal weights $\Delta_{r,s}^-$ of the simple modules $\mathcal{X}_{r,s}^-$ are not roots of f .

Open problems

- Construct projective covers of all simple modules.
- Compute fusion tensor product on $\mathcal{M}_{p_+, p_-}$ -mod.
- Classify modular invariants.
- Classify conformal field theories with $\mathcal{M}_{p_+, p_-}$ symmetry.

The End

Thank you for your attention!